Signal Detection and Metacognition

Sensitivity, bias and confidence, applied to LLM introspection

---

Measuring AI Introspection

Overview

Part 1: Sensitivity and saying yes

1. The running example
2. Four outcomes
3. Two bell curves and a line
4. Same hit rate, different observers
5. Sensitivity and criterion as numbers

Part 2: Confidence about one's own answers

6. Type-1 and type-2 tasks
7. The confidence-accuracy table
8. Sensitivity, bias and efficiency
9. Correlations and type-2 d'
10. AUROC2
11. Meta-d' and the M-ratio

Part 3: Probabilities and awareness

12. The Brier decomposition
13. Three senses of calibration
14. Michel's four conditions

Part 4: Language models

15. Concept injection as a type-1 task
16. Injection moves the criterion
17. A logit as a decision variable
18. Confidence papers as type-2 tasks
19. A probe as the ideal observer

Part 5: Limits

20. Where the analogy stops
21. Summary
22. Reading

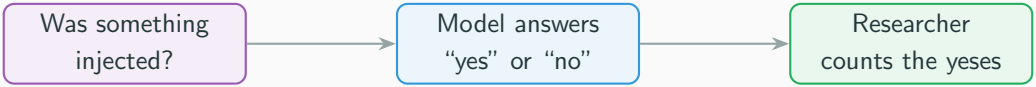
Part 1

Sensitivity and saying yes

The running example

Lindsey (2025) runs a *concept injection* experiment:

1. Take the model's internal activation pattern for a word, such as *bread*.
2. Add it to the model's activations on some trials, and leave the activations alone on others.
3. Ask: *Do you detect an injected thought?*



Signal detection theory is the standard way to read a pile of yes/no answers. It splits a rate of "yes" into two things: how well the model tells injection from no injection, and how willing it is to say yes.

Four outcomes

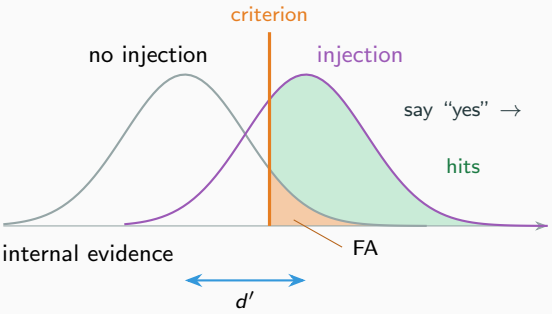
Every trial lands in one of four cells:

	Model says "yes"	**Model says "no"**
Injected	**Hit** *noticed a real signal*	**Miss** *missed a real signal*
Not injected	**False alarm** *"yes" to nothing*	**Correct rejection** *"no" to nothing*

The **hit rate** H is the share of "yes" on injected trials. The **false-alarm rate** F is the share of "yes" on trials with nothing injected.

Each row needs its own trials, and both rows are needed.

Two bell curves and a line



On each trial the model has some amount of internal evidence. It varies from trial to trial, so each condition gives a bell curve.

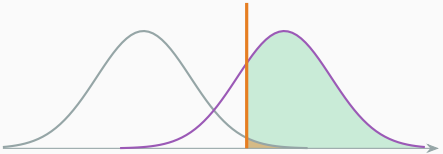
Sensitivity (d'): how far apart the two curves are.

Criterion: the line. Evidence to the right of it gets a "yes".

Green area: hits. Orange area: false alarms (FA).

Same hit rate, different observers

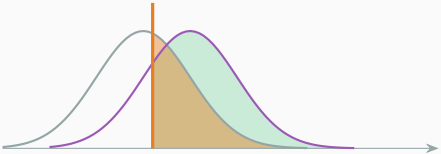
High sensitivity, strict criterion



Curves far apart; line far right.

Almost no false alarms.

Low sensitivity, lax criterion



Curves overlap; line far left.

Many false alarms (orange).

Both observers say "yes" on the same share of injected trials. One is sharp and cautious; the other barely tells the conditions apart and says yes readily.

A hit rate alone cannot tell them apart. Only the false-alarm rate can.

Sensitivity and criterion as numbers

Under the equal-variance Gaussian model:

$$d' = z(H) - z(F) \qquad\qquad c = -\tfrac{1}{2}\left[z(H) + z(F)\right]$$

z turns a rate into a position on the bell curve, measured in standard deviations (the inverse of the standard normal cumulative distribution function).

d': distance between the curves
stays the same if only the line moves

c: position of the line
the tendency to say yes

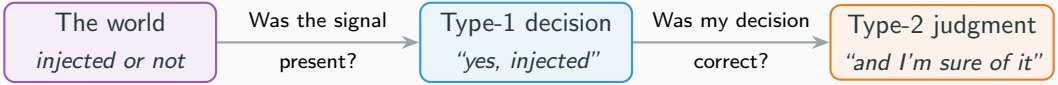
Percentage correct mixes the two. d' stays constant when only the criterion changes (Fleming & Lau 2014).

Part 2

Confidence about one's own answers

Type-1 and type-2 tasks

Metacognition research applies the same split one level up, to confidence.



A **type-1** task asks about the world. A **type-2** task asks about the observer's own type-1 decision.

The type-2 score is checked against whether the observer was right, not against the state of the world (Fleming & Lau 2014).

The confidence-accuracy table

Same four cells as before, one level up. Rows are now *was I right*, columns are *how sure was I*:

	High confidence	**Low confidence**
Correct	**Type-2 hit** *right and knew it*	**Type-2 miss** *right but unsure*
Incorrect	**Type-2 false alarm** *wrong but sure*	**Type-2 correct rejection** *wrong and unsure*

Good metacognition fills the green cells: confidence is high when the decision was right and low when it was wrong.

Sensitivity, bias and efficiency

Three quantities are defined on this table (Fleming & Lau 2014):

Sensitivity
Does confidence separate my right answers from my wrong ones?

Bias
How confident am I overall, right or wrong?

Efficiency
Sensitivity relative to how well I do the task itself

Bias is also called over- or underconfidence, or calibration. Sensitivity is also called metacognitive accuracy or discrimination.

A subject can be confident overall and still have poor sensitivity, if confidence is equally high on correct and incorrect trials.

Correlations and type-2 d'

Fleming & Lau 2014 review the obvious measures of metacognitive sensitivity. Two families fail:

Phi and gamma

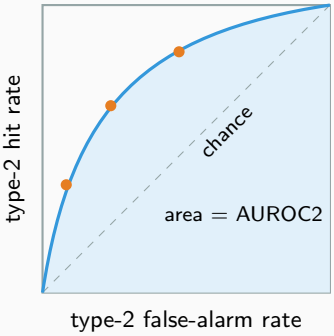
Correlate confidence with accuracy. Both move with bias: a modest subject who rates every trial low gets a low correlation, for reasons that have nothing to do with sensitivity.

Type-2 d'

Runs the d' formula on the type-2 table. That assumes equal-variance bell curves for correct and incorrect trials. Galvin et al. (2003) showed they are unequal and non-Gaussian, so the measure is confounded with bias.

Both fail the same way: a change in how confident the subject is overall shows up as a change in sensitivity.

AUROC2



Treat each confidence level as a criterion. Each one gives a point: a type-2 hit rate and a type-2 false-alarm rate (orange dots).

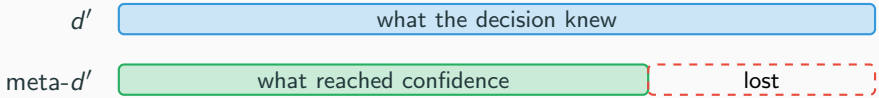
Join the points into a curve. The area under it is **AUROC2**. The diagonal is confidence that says nothing about correctness.

Good: free of bias.

Problem: it rises with task performance. The same subject scores higher on an easy task than on a hard one.

Meta-d' and the M-ratio

Meta-d' (Maniscalco & Lau 2012): the d' that an ideal observer, one who uses all the evidence, would need to produce the confidence performance actually seen. Same units as d', so the two compare directly.



M-ratio $=$ meta-d'/d': how much of what you know reaches your confidence.

M-ratio $= 1$: a metacognitively ideal observer; all the evidence reached confidence.

M-ratio $= 0.7$: 30% of the evidence was lost on the way to confidence (Fleming & Lau 2014).

Michel (2023) calls the M-ratio the gold standard for metacognitive efficiency. When d' is small the ratio is unstable, and meta-$d' - d'$ or the log ratio is used instead.

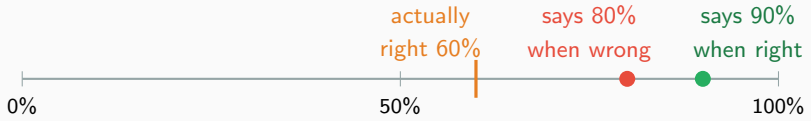
Part 3

Probabilities and awareness

The Brier decomposition

A stated probability gives bias an absolute scale: a stated 70% can be checked against the proportion correct. The Brier score, the mean squared gap between stated probability and outcome, splits as

$$PS = \underbrace{O}_{\text{outcome variance}} + \underbrace{C}_{\text{calibration}} - \underbrace{R}_{\text{resolution}}$$



Calibration (bias): poor; both confidence levels sit far above 60%. **Resolution** (sensitivity): good. Right answers always get more confidence than wrong ones (Fleming & Lau 2014).

Because O depends on accuracy, the Brier score moves with task performance.

Three senses of calibration

Papers on LLM introspection use "calibration" in more than one sense:

Sense	Meaning	Where it appears
Metrological	Relating an instrument's indications to values of the measurand, against a standard	Metrology
Probability	Stated confidence agrees with proportion correct; the C term of the Brier score	Kadavath et al.'s expected calibration error; Ferrara's Brier score
Discriminatory	Confidence rank-orders outcomes; metacognitive sensitivity under another name	Moran and Whiting, Def. 3.1

Kadavath et al. (2022) state the difference between the second and third: the area under the curve measures discrimination and ignores calibration.

Michel's four conditions

Michel (2023) separates **validity** (does the procedure measure the intended attribute?) from **accuracy** (how well?), and states four conditions:

Condition	Content	Guarantees
Consciousness-selectivity	Unconscious sensory activity is a source of metacognitive inefficiency	**Validity**
Optimal metacognition	Metacognition is optimal for decisions based on conscious activity	**Accuracy**
No overflow	All conscious activity is available for metacognitive decisions	**Accuracy**
No hallucination	Activity driving false alarms is not conscious	**Accuracy**

Other sources of inefficiency are confounds. Controls: remove them (*ceteris absentibus*), equate them across conditions (*ceteris paribus*), or make them negligible (*ceteris neglectis*).

Part 4

Language models

Concept injection as a type-1 task

The stimulus is internal, but the question is type-1: was a signal present?

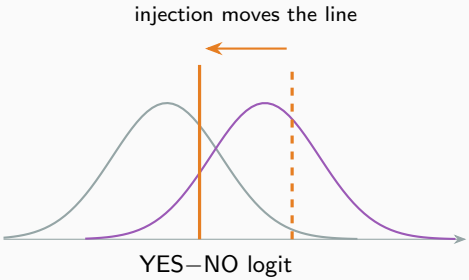
Hit rate	False-alarm rate
"yes" on injection trials	*"yes" with nothing injected*

Lindsey (2025) reports **no false positives in 100 control trials**. That is good news for the model and a problem for the arithmetic:

$$F = 0 \;\Rightarrow\; z(F) = -\infty \;\Rightarrow\; d' = z(H) - z(F) \text{ is infinite.}$$

So d' depends on the **correction chosen for the empty cell**. Ferrara (2026) uses a $1/(2N)$ correction against a paired sham trial and reports a pooled $d' = 0.0039$.

Injection moves the criterion



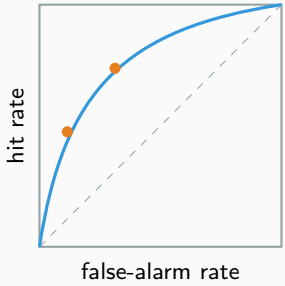
Hahami et al. (2025) read the YES-minus-NO logit in Llama-3.1-8B-Instruct. They apply the same injection to control questions whose answer is no.

The rise on the detection question and the rise on the controls correlate at $\mathbf{r = 0.999}$.

In signal detection terms, the injection shifts the **criterion**: the model becomes more likely to say yes to any question.

Lindsey's check on unrelated questions in Claude models finds no increase in yes answers.

A logit as a decision variable



Rating scale: a few criteria, so a few points on the ROC curve (orange dots).

Logit difference: continuous. Every possible threshold is a criterion, so the full receiver operating characteristic curve (blue) comes from one run, without asking for ratings.

Hahami et al. read every answer this way.

Confidence papers as type-2 tasks

Three papers score a model's confidence against its own correctness:

Kadavath et al. 2022
Probability calibration and AUROC

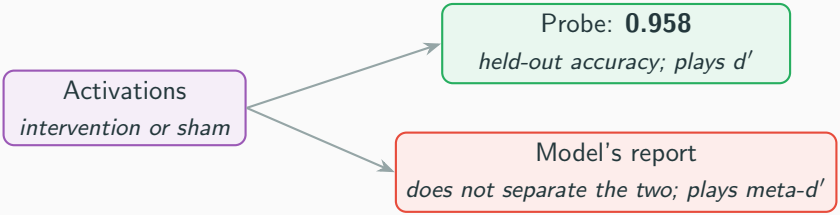
Ackerman 2025
Partial correlation between delegation and correctness: the phi family

Moran & Whiting 2026
Thresholded latent normal variable, one threshold per model

Ackerman's measure is in the phi family, which moves with bias. Moran and Whiting's threshold separates a model's yes rate from the shared signal; they state that the normality assumption is presumed rather than tested.

A probe as the ideal observer

Meta-d' compares confidence with the evidence the decision had. The closest LLM version compares the model's report with a probe trained on the same activations.



Ferrara (2026), Qwen2.5-7B-Instruct. Ferrara defines the ratio of report margin to probe margin as an M-ratio analog.

In plain terms: the information is in the activations, and it does not reach the report.

Part 5

Limits

Where the analogy stops

- **Items, not internal noise.** Signal detection's standard design holds the stimulus constant, so variation in answers reflects noise inside the observer (Fleming & Lau 2014). Many LLM experiments sample at temperature 0, where an item always gets the same answer. Variation then comes from items, and d' describes distributions over items.
- **One subject.** Human studies estimate efficiency per subject and compare subjects. A model sampled many times is one subject, and differences between models mix architecture, training and prompt format.
- **Type-1 and type-2 can merge.** Meta-d' rests on a model of the type-1 decision. In an LLM the confidence prompt can make the model solve the item again. Moran and Whiting report that reasoning-trained models often do, which turns a type-2 judgment into a second type-1 attempt.

Summary

Idea	Question for an introspection experiment	Card stage
Hits and false alarms	Is the yes rate without injection reported?	Instrument
d' versus criterion	Is a yes-bias separated from sensitivity?	Measurement model
Empty cells	Which correction is used when $F = 0$?	Uncertainty
Type-1 versus type-2	Is the task about the world or the model's answer?	Measurement model
Sensitivity, bias, efficiency	Which of the three does the measure track?	Measurement model
AUROC2 and Brier score	Does the number move with task performance?	Uncertainty
Three calibrations	Which sense of calibration is meant?	Calibration
Meta-d' and the probe	Is the report compared with a probe?	Comparability
Constant stimuli	Does variation come from items or the model?	Validity conditions

Reading

- Fleming & Lau, How to measure metacognition (2014): sensitivity, bias and efficiency; bias-free measures; the Brier decomposition.
- Maniscalco & Lau, A signal detection theoretic approach for estimating metacognitive sensitivity from confidence ratings (2012): meta-d'.
- Maniscalco, type-2 SDT code: fitting meta-d'.
- Michel, Confidence in consciousness research (2023): validity versus accuracy; conditions and confounds.
- Michel, The old and new criterion problems (2023): response bias in type-1 reports.
- Signal detection and metacognition: the site page this deck follows.